BIOMASS AND BIOENERGY 35 (2011) 2435–2443

Available at www.sciencedirect.com

ScienceDirect

http://www.elsevier.com/locate/biombioe



Spatial scale and social impacts of biofuel production

Dan van der Horst [a,*], *Saskia Vermeylen* [b]

[a] *School of Geography, Earth & Environmental Sciences, University of Birmingham, United Kingdom*
[b] *Lancaster Environment Centre, Lancaster University, United Kingdom*

ARTICLE INFO

Article history:
Received 28 March 2008
Received in revised form
1 November 2010
Accepted 4 November 2010
Available online 8 December 2010

Keywords:
Ethical biofuels
Scale of production
Environmental governance
Multifunctional crops
Livelihoods

ABSTRACT

The prospect of biofuels going 'mainstream' has drawn more attention to the social impacts of the production and use of transport biofuels. Since 2007, many media stories have appeared about alleged negative impacts of biofuels, notably the price of food going up or land-grab by plantation developers. These stories stand in stark contrast with the rosy picture painted by some academics involved in the technical development of bio-ethanol or biodiesel. This paper explores the questions when and why negative social impacts are likely to occur and under what circumstances more positive impacts might be expected. These impacts are discussed for three geographically defined biofuel supply chains; north–north, south–north and south–south. These three systems differ in the spatial scale of production and consumption and with that comes a different distribution of environmental, social and economic impacts. In the case of domestic production and consumption in developed countries, the social impacts are relatively minor and can be mitigated by social policies. Large scale, export-oriented production systems in developing countries could theoretically yield positive social impacts, but this would require on the one hand the tailored design of 'pro-poor' social innovations and interventions on the ground and on the other hand a certification of the supply chain feeding into consumer demand for 'ethical' fuel. The latent existence of this demand might be significant but recent NGO campaigns have severely undermined the ethical credentials of biofuels. It would require a persistent and collaborative effort to restore the brand value of 'green' fuel, an effort which will require better legislation and radically improved monitoring and enforcement practices in countries where the very absence of these has led to, and is still causing, the large scale destruction of habitats that are carbon sinks of global importance. The significant levels of government funding for biofuels stand in strong contrast with the problematic environmental and social governance of international biofuels supply chains. Notwithstanding the 'must tackle climate change' rhetoric by policy makers and in policy documents, this suggests that biofuels policy may be primarily driven by other concerns, especially regarding energy security. We argue that policies that are designed for a rather narrowly defined purpose of 'security of supply', cannot be realistically expected to yield high social or environmental benefits, and certainly not abroad.

© 2010 Elsevier Ltd. All rights reserved.

* *Corresponding author.*
E-mail addresses: D.vanderhorst@bham.ac.uk (D. van der Horst), S.vermeylen@lancaster.ac.uk (S. Vermeylen).
0961-9534/$ – see front matter © 2010 Elsevier Ltd. All rights reserved.
doi:10.1016/j.biombioe.2010.11.029

1. Introduction

Since the rapid increase in oil price in the period 2005–2008, the era of cheap oil appears to have come to an end. This has transformed the economic prospects of alternative transport fuels, including that of liquid biofuels like bioethanol and biodiesel. This has gone hand in hand with the development in many countries of national policies to boost the production and use of renewable energy, including liquid biofuels. These policies are usually justified by a mixture of environmental, social, developmental/economic and security of supply arguments. Whilst in principle biomass energy has the potential to fulfil each of these objectives, it is not so straight-forward to achieve them all simultaneously: in practice the choice of specific policy designs and project types often privileges the achievement of one policy objective at the expense of another [1]. This paper explores how the social impacts of biofuel production may be at odds with the push to increase the production of liquid biofuels as global commodities.

Proponents of liquid biofuels, including academics who have a strong professional interest in particular plant species or energy technologies, talk about a range of (potential) social benefits. Some hail the multifunctionality of Jatropha [2] or even wax lyrically about the "life promoting energy" of the oil palm [3]. The key argument is that of employment and rural development, but additional supporting arguments include the prevention of soil erosion, the possibility of intercropping, the provision of animal fodder, the use of parts of plants for building materials, medicines etc.

These optimistic views contrast strongly with recent reports by NGOs and the media about poor people suffering from the price increases of staple foods as a result of food crops being converted to biofuels, rainforest being cut down to make way for oil palm plantations, indigenous or minority communities being killed or chased off their land so that it can be turned into energy crop plantations [1] [4,5].

It may be tempting to undermine the poignancy of these diverse stories from Latin America, Africa and South East Asia by pointing at the flaws in some of the arguments put forward through simplistic media headlines. Price increases in food are clearly influenced by a combination of factors, including the growing consumer demand in China and India, the impacts of severe weather on crop yield (e.g. the draught in Australia in recent years) and the rising cost of fossil fuel which increases the costs of mechanised agriculture, the use of synthetic fertilisers and the transportation of food to the consumer. The destruction of rainforests may involve logging companies that (so far) have had no interest in biofuels, and it is the lack of good governance and effective policing that allows it to take place in some countries. Similarly, issues of land-grab could be portrayed as individual cases that have no generic applicability across the world, that are related to historical problems in specific countries and that are certainly not limited to the production of biofuels.

Regardless of the caveats in these media stories, there is a world of difference between the 'promise' of the benefits that liquid biofuels could bring, and what actually happens 'on the ground' – where the production of biofuels can have very negative social impacts. It could thus be argued that the justification for continuing public policies to promote transport biofuels, hinges to a large extent on the question of how to avoid these negative social impacts, and (ideally) how to obtain positive social impacts in stead. This in turn requires a better understanding of the nature of negative or positive social impacts in relation to the different types of possible policy interventions and different types of (potential) biomass energy systems. This paper aims to contribute to this understanding by exploring under what generic circumstances the production of liquid biofuels could be expected to yield positive social impacts.

Firstly this paper will draw attention to the main existing approach of assessing the social impacts of policy interventions and highlight it's relevance with respect to liquid biomass energy projects. As a next step, the potential social impacts of the production of liquid biofuels are discussed for developed and developing countries respectively. With regards to the latter, we draw on insights from environmental economics to explore to what extent the reported negative social impacts are inherent in the way in which liquid biofuels are produced. In the subsequent discussion we explore the motivations behind the design of and support for existing liquid biofuel systems.

2. What are social impacts

The development of Social Impact Assessment (SIA) guidelines was a logical follow-up to the development of Environmental Impact Assessment (EIA). EIA entered US legislation in the US through the 1970 National Environmental Policy Act (NEPA). In 1992, a group of US social scientists formed the Interorganizational Committee on Guidelines and Principles for Social Impact Assessment (IOCGP). Consisting of professional sociologists, anthropologists, social psychologists, geographers, land use planners, economists, natural resource social scientists and landscape architects, this group first published their 'principles and guidelines' in 1993 (see Table 1). The IOCGP define social impacts as "the consequences to human populations of any public or private actions that alter the way in which people live, work, play,

[1] The following sample illustrates the level of media and NGO attention for this issue. BBC news website (www.bbc.co.uk) (last accessed 27/10/2010)

- 3 January 2007. Power station drops palm oil idea
- 1 February 2007. Mexicans stage tortilla protest
- 2 April 2007: EU fuel push 'might damage forests'
- 12 April 2007. Deaths in Uganda forest protest.
- 11 July 2007. Plea over Indonesia palm oil plan
- 20 July 2007. Food prices on the rise and rise
- August 2007. Losing land to palm oil in Kalimantan

Examples of reports by environmental NGOs include (last accessed 27/10/2010):

- Greenpeace UK. Biofuels: green dream or climate change nightmare? 9 May 2007http://www.greenpeace.org.uk/blog/climate/biofuels-green-dream-or-climate-change-nightmare-20070509
- Friends of the Earth The oil for ape scandal. 2005. http://www.foe.co.uk/resource/reports/oil_for_ape_full.pdf
- African Biodiversity Network. Agrofuels in Africa. The impacts on land, food and forests. 2007. http://www.africanbiodiversity.org/media/1210585739.pdf

Table 1 – The principles and guidelines of Social Impact Assessment developed by the IOCGP.

1. *Achieve extensive understanding of local and regional settings to be affected by the action or policy*:
- Identify and describe interested and affected stakeholders and other parties
- Develop baseline information of local and regional communities

2. *Focus on key elements of the human environment*
- Identify the key social and cultural issues related to the action or policy from the community and stakeholder profiles
- Select social and cultural variables which measure and explain the issues identified

3. *Identify research methods, assumptions and significance*
- Research methods should be holistic in scope, i.e. they should describe all aspects of social impacts related to the action or policy
- Research methods must describe cumulative social effects related to the action or policy
- Ensure that methods and assumptions are transparent and replicable
- Select forms and levels of data collection analysis which are appropriate to the significance of the action or policy

4. *Provide quality information for use in decision making*
- Collect qualitative and quantitative social, economic and cultural data sufficient to usefully describe and analyze all reasonable alternatives to the action
- Ensure that the data collection methods and forms of analysis are scientifically robust
- Ensure the integrity of collected data

5. *Ensure that any environmental justice issues are fully described and analyzed*:
- Ensure that research methods, data and analysis consider underrepresented and vulnerable stakeholders and populations
- Consider the distribution of all impacts (whether social, economic, air quality, noise or potential health effects) to different social groups (including ethnic/racial and income groups)

6. *Undertake evaluation/monitoring and mitigation*
- Establish mechanisms for evaluation and monitoring of the action, policy or program
- Where mitigation of impacts may be required, provide a mechanism and plan for assuring effective mitigation takes place
- Identify data gaps and plan for filling these data needs

relate to one-another, organise to meet their needs and generally cope as members of society. The term also includes cultural impacts involving change to the norms, values, and beliefs that guide and rationalise their cognition of themselves and their society" [6]. Other organisations have followed in the footsteps of the IOCGP and developed broadly similar approaches. For example the World Bank has developed guidance for carrying out Social Analysis [7] and Poverty and Social Impact Analysis (PSIA) [8] to inform and evaluate their own investment decisions. International trends in SIA are now moving the focus towards building social capital, capacity building, good governance, community engagement and social inclusion [9].

A closer look at the principles of the SIA guidelines (Table 1) reveals that they are more easily implemented in a social context that is familiar, and in a culture of governance that aims to be more inclusive and seeks to ensure the participation of all groups in the policy process, including weaker or more vulnerable groups in society such as the poor, women and children, people with disabilities, indigenous people or other minorities. As a consequence, it can be argued that SIA guidelines are more easily implemented, and negative social impacts are more easily avoided or mitigated, in a domestic project in a developed country with a strong democratic and social welfare tradition, than in a project that causes social impacts beyond national boundaries or that takes place in locations where governance structures are weak or less concerned about the fate of vulnerable groups. This difference in the likelihood of avoiding or mitigating social impacts, can be further explored by splitting liquid biofuel systems up into different categories on the basis of the scale and location of production and consumption. This is a categorization on the basis of different spatial scales of cause and effect. The cause in this case is the demand for biofuel which is created through state intervention (targets, tax breaks etc) in the country of consumption, while most of the social effects take place at the place of production. In the following sections we will discuss the expected social impacts of large scale production systems in developed and developing countries respectively.

3. Social impacts of liquid biofuel production in developed countries

Many developed countries have developed national policies that are aimed at increasing the domestic production of liquid biofuels. Such domestic policies result in direct social impacts within the country only, but indirect impacts may be felt across the border. The main indirect impact is that on world commodity prices. If countries like France and Germany decide to increase production of biodiesel from home grown oil seed rape, this will 'capture' some agricultural land that is currently used to grow other commodities and/or it may capture some of the oil seed rape that is currently produced for food, cosmetics etc. A domestic liquid biofuel policy can thus result in a reduced supply of certain commodities on the world market. This will raise the price of the commodity and thus hit poor consumers abroad and at home (as has happened with the US corn to ethanol programme). As a result of increased world prices, more of the displaced commodities may be grown in other countries, including developing countries. The impacts of large scale production of biofuels in or on developing countries will be discussed in the next section. The remainder of this section will explore what the direct (domestic) social impacts may be of such policies.

Rural communities are usually characterised by relatively high levels of social capital, but relatively low levels of economic opportunity. The loss of a few jobs, and the lack of prospect to find new ones, results in outward migration which can tip communities over certain thresholds, resulting in the closure of the local shop, bank or school. The processing of energy crops and production of transport biofuels in rural areas could provide new jobs, retain more money in the rural economy and support local services. However the trade-off is that this implies a decentralised system of small plants, and thus runs counter to the logic of efficiency, both in economic and in energy/carbon terms. Further research would be

required to study the environmental, economic and social trade-offs implicit in the choice of location and scale of liquid biofuel processing plants.

Clearly the rural poor could potentially benefit from cheap fuel as they are much more car dependent to gain access to basic services and as they have to spend a disproportionately large amount of their meagre income on transport. However the authors did not come across liquid biofuel projects which aimed to provide cheaper fuel to the rural poor.

Apart from the cost of food, direct social impacts are comparatively minor in domestically produced liquid biofuels in developed countries since most of the stakeholders involved are not particularly vulnerable. Arguments of rural development are often used to support such policies, but although farming is perceived to be in crisis in many developed countries, questions can be raised about the importance of these types of farming subsidy to the rural economy. Farming these days is certainly no longer the mainstay of the rural economy of the European countries that offer the strongest financial support for biofuel production; it has long been surpassed by sectors such as tourism or the services industry. In parts of Eastern Europe farming is still an important provider of jobs in rural areas, but these are middle income countries that have chosen to direct very little of their (more limited) tax money to subsidise the production of liquid biofuels. If 'rural development' is simply read as a euphemism for supporting the farmers, then subsidies for the production of energy crops can be seen as just the latest initiative in the state's ongoing effort to maintain a way of life that is threatened. It would be beyond the scope of this paper to discuss the transformations of farming as a culture in increasingly post industrial societies (highly urbanised with economies dominated by the service sector), but it should be clear from the level of farming subsidies in most developed countries since the Second World War that the strength of the farming lobby often far exceeds the size of the community it seeks to serve. It could be argued that this historic trend of farm support is justified by the view of a significant sections of the population, of farmers as the stewards of traditional landscapes and regional foods who maintain society's link with its agricultural and pre-industrial roots, but it is far from clear how the switch from the production of food to the production of new automotive fuels will chime with such romantic public sentiments of farming.

4. Social impacts of large scale liquid biofuel production in developing countries

This section will focus on three, somewhat interrelated, themes in which the picture presented by the proponents of liquid biofuels tends to differ from the situation on the ground as it is reported by NGOs and the media. The first theme relates to the question of what land will be utilised to provide this increase in production. The second theme relates to the question about the distribution over different sections of society, of different types of benefits (or ecosystem services) arising from a particular type of land use. The third and last theme explores the impact of the large scale introduction of cash crops on the livelihoods of rural communities.

4.1. *Why targeting of 'marginal' land comes at the expense of marginal communities*

In developed countries, land of marginal agricultural productivity is often appreciated for its landscape or biodiversity value, especially as these values are scarce on more productive land which has been transformed by large-scale mechanised and intensive agriculture. In developing countries, widespread poverty means that such non-productivist sentiments are less prevalent. Land of marginal agricultural productivity is often viewed by developers as 'cheap' and therefore attractive for conversion into biofuel plantations. However this 'waste' land is rarely uninhabited or unused by the people who live there. As a rule, the more marginal their livelihoods are, the more likely rural people will depend on the land for their day-to-day struggle for survival. The land will yield fuel, medicines, wild food, building materials etc. to people who do not have the means to obtain equivalent (or better quality) goods or services in the formal (money) economy. A government's definition of 'degraded' or 'waste' land is perhaps informed by the land's previous productivity or by the current absence of agricultural systems that produce commodities for the world market, i.e. bring in foreign currency and/or tax revenue. This remote view from the centre of power, is often at odds with the views of the local people.

Countries where social and environmental governance is weak, often attract investments by overseas companies eager to reduce their production cost by avoiding the need to comply with social and environmental standards. This phenomenon is known as 'pollution havens' [10,11], and it has some relevance for the biofuels sector. Indeed when a government is weak or corrupt, marginal agricultural land may not be the only category of land that is likely to be targeted by powerful business groups: The conversion of forested land can be even 'cheaper' for these companies because of the revenue that can be generated from logging or because the land is thinly occupied by poor, minority or indigenous groups that are politically weak and have no formal title (i.e. registered in state records) to the land, and can thus be more easily displaced. It is worth noting that this point is equally relevant for second generation biofuels. In fact, if cellulose could be cheaply converted into automotive fuel, this would turn all biomass into a bulk commodity and encourage clear-felling. The local social and environmental impacts of this level of slash (in the south) and burn (in the north) could be huge.

The issue of 'previous land cover' which is critical for the environmental performance of a biomass energy system (as measured in an LCA), is thus often closely linked to the issue of 'previous land ownership' which is critical for the social performance of the biomass energy system. Furthermore, displaced people have to go somewhere. There are many instances where landless poor are at the settlement frontier, entering remaining forests, the land of (other) indigenous peoples or nature reserves in a desperate effort to sustain themselves and their families. There are large areas of deforested and degraded land that are now used for extensive cattle grazing by relatively affluent cattle farmers and which could be made more productive by planting energy crops. But whether these farmers sell the land or switch to energy crops,

this too can trigger a displacement effect as it may be enticing cattle farmers to move into 'new' areas and thus increase the pressure on existing forests. As far as we are aware, these displacement effects are yet to be included in any LCA study of biofuels. It should be clear that the potential environmental (and social) implications of these displacement effects are likely to be significant and could potentially be huge. This means that the drawing of system boundaries in LCA studies of biofuels is implicitly value-laden [12] and potentially highly controversial. Considering the scientific knowledge and technical skills required for LCA, it is understandable that most LCA experts would prefer to draw fairly narrow system boundaries around their LCA study. Accounting for these displacement effects would require an approach that is much more interdisciplinary and multi-method.

Even in occasions where the land appears to be unoccupied by squatters or informal settlements, it is important to understand the history. 'Empty land' (*terra nullius*) is almost invariably a myth [13]. Nomadic people may provide seasonal occupation, or local people may have been forced to move out as a result of external pressures such as discriminatory government policies, violent conflict or the collapse of traditional livelihoods. It may for example be that subsidised agricultural exports from industrialised countries have made local agriculture economically unsustainable and people have had to give up farming and move to towns in search of alternative livelihood strategies. These people still retain a strong sense of rural identity and sense of ownership and belonging with regards to their ancestral land. Global liquid biofuel demand results in increased cost of food and this will further marginalise the urban poor and impact on the nature of urban–rural dependencies in developing countries. The livelihood strategies of the poor are often heavily depend on mechanisms of sharing and support within the extended family, clan or village. Often young adults migrate to the cities in search for work whilst part of the family remains in the rural area of origin. The ties are maintained however and people and resources (food, money) are transferred in support of the most needy within the family [14]. Increased food insecurity in cities is thus likely to send the urban poor back to their rural areas of origin or make them more dependent on shipments of food from their rural relatives. In short, expensive urban food will increased dependency on food produced in rural areas and will increase the pressure and demand for land.

4.2. *Why extraction of commodities by large corporations tends to come at the expense of local communities*

Environmental economic theory explains environmental damage as 'externalities' that are caused by market failure, i.e. the inability of the market to take certain environmental issues into consideration. Government intervention is deemed necessary to 'internalise' these environmental costs, for example by taxing polluters. Setting appropriate taxation levels requires an estimation of the damage in monetary terms. Environmental economists speak of the total economic value (TEV) of an organism or a piece of land as the sum of all market and non-market values, the latter being further sub-divided into use and non-use values. Market values are the monetary values of a commodity that is sold and bought on the market. Non-market use values relate to things that are harvested and consumed without being traded on the market. Examples may include the gathering of wild foods or other materials for direct use or home consumption. Non-use values relate to things that are enjoyed by people without being diminished as a direct consequence of this enjoyment. Examples include protection against natural hazards (e.g. soil erosion, avalanches, flooding) but also amenities such as beautiful views, the sound of flowing water, or even spiritual or religious associations with (aspects of) the natural environment. Environmental economists have developed a suite of tools to estimate the monetary value of these non-market values. Utilising these tools, studies consistently show that the TEV of (parts or aspects of) nature vastly exceed its market value [15–17].

Emerging from this is the message that natural resource management decisions that are mainly informed by market values, are likely to result in wasteful and damaging practices when we consider the range of different values or the Total Economic Value of the resource.

This is of relevance to the situation in developing countries where liquid biofuels are (to be) produced for the global market. Clearly large companies involved in this business depend for their profit on the extent to which they can extract a market value from the energy crops or the land on which they are grown. These companies are unlikely to benefit from the non-market values and will therefore not be particularly interested in providing or safeguarding such values. Indeed it would require pressure from powerful stakeholders to entice such a company to forsake some of the market value it can extract, in order to safeguard the provision of other values or ecosystem services. If at all existent, such pressure tends to come from affluent consumers, influential NGOs or from national governments with strong green and social credentials. This means that companies that are operating in developing countries (where state governance structures generally tend to be weaker, less democratically inclusive and/or less focused on the environment) and are extracting commodities for consumption in developed countries are less likely to feel pressure to improve their environmental or social performance, especially if these commodities are not normally or easily branded and traced. For biofuels branding and tracing is particularly difficult as in a large scale biodiesel production chain, oil from different origins may be collected for bulk shipment and storage, it then undergoes chemical treatment (trans-esterification) and is then blended again as a small minority ingredient (often 5%) of 'normal' diesel that is sold at the petrol station. Whereas for items such as food (e.g. fair trade, organic, catch not exceeding sustainable yield), textiles (e.g. no child labour) or cosmetic (e.g. not tested on animals) there are widely known and felt concerns amongst affluent consumers [18], this same concept is yet to be established for biofuels. The stories in the media about the negative impacts of growing liquid biofuels, are likely to make consumers more jittery and reluctant to buy into such an 'unethical' product.

In this context, questions about the social impacts of liquid biofuel, are essentially questions about the distributional consequences of the production system, i.e. to what extent are the various use- and non-use values benefiting (or likely to

benefit) different sections of society? It is highly likely that many of the non-use values are specifically benefiting local communities (e.g. protection against soil erosion, gathering of wild foods, maintenance of places of worship) or future generations (carbon sequestration, protection of biodiversity). The economic logic that drives investment decisions in commercial companies clearly favours large-scale mono-culture plantations (and in all likelihood also) the use of pesticides and GM varieties which would not be acceptable for food crops, so as to maximise yields in the short term. This production model, imposed by a powerful stakeholder (the company) is a particularly poor deliverer of non-market goods and services to local communities or future generations, who lack the power to block these developments or negotiate a fair compensation.

4.3. *Why the switch to cash crops, and the creation of paid jobs, may not be a net gain for rural communities*

Job creation and income generation for the rural poor is a key argument used by proponents of transport biofuels. However questions can be raised about the suggestion that large scale biofuel plantations provide a net gain to rural communities. Unless they have formal title to their land or significant numbers of cattle, rural communities in developing countries appear to have limited material assets or access to cash generating activities. This outside perspective of poverty is one-sided however. Financial transactions measure only a limited aspect of how rural communities live, survive or thrive. The fact that people are willing to engage in a particular income-generating activity, should not automatically be seen as evidence of positive social impacts. This willingness must be assessed against alternative livelihood options that are realistically available to the person. It may well be that the person suffers from deprivation caused by earlier exploitation which is associated with the current 'opportunity' for income generation. Classical examples for that include landless poor working for powerful estate owners. Indeed instances of slavery are still far too common on Brazilian sugarcane estates. [2]

The status of the rural 'poor' is better assessed by using a sustainable livelihoods approach, which identifies five different types of capital (human, financial, natural, social and physical capital) which underpin their livelihood strategies [19]. Long established and indigenous rural communities are experienced in living in a highly variable or seasonal environment and their traditional methods of survival are based on managing this risk rather than on optimising their income. And even if/when they earn some extra money, they rarely have access to financial services which would allow them to save or accumulate it. As mentioned earlier, truly 'empty' land that is suitable to sustain any kind of human activity or inhabitation is very rare indeed so that new biofuel plantations are invariably planted on land previously utilised by local communities in one way or another. A voluntary or forced switch from subsistence farming to the farming of non-food cash crops brings with it an increased dependency on world prices. This may be fine when the prices of the cash crop exceed the price of food that needs to be bought, but when prices of the cash crop drop or world food prices go up (as has been the case in recent years), then this will reduce food security. Whereas the weather is often the main threat to the food security of subsistence farmers, the switch to cash crops brings with it additional and external threats such as policies in developed countries, extreme weather in other parts of the world and conflicts or financial market turmoil that originate in far-away areas. A stark illustration of this was provided by the Icelandic ash cloud over Europe in the spring of 2010; It did great economic damage to the commercial flower farmers in Kenya, who were stuck with a highly perishable product for which there was no local demand.

The involvement of rural communities in the production of liquid biofuels cannot be evaluated through simplistic proxies such as the number of jobs on the plantation or the average pay per worker. What is required is a much more detailed analysis of how the livelihood strategies and outcomes of rural communities and the individuals within these are transformed by the changes in land ownership, land management and land use associated with the switch towards the production of biofuel. Whilst it may be true that oil palm was traditionally used to provide food, medicine, woven material or wine [20] and that Jatropha has been used to provide medicine, tannins, live fencing, honey or erosion protection [21], it is important to assert through in-depth fieldwork to what extent these, or other (and possibly new) functions or services are provided by the plant and the local agricultural system in which it is (often quite recently) integrated. A case study of oil palm plantations and adjacent (and associated) oil palm producing smallholders in Papua New Guinea by Koczberski and Curry [22] provides a good example of the complexity on the ground. Their research is highly critical of World Bank advice to the government to increase the faltering productivity of oil palms grown by the smallholders. They argue that to view of smallholders as solely oil palm producers (as the World Bank and the palm oil industry tended to do) is limiting the range of potential interventions to increase production and overcome other farm management problems such as the reluctance to poison and replant old palms. By showing that this reluctance stems from the high price of pesticide, the need to purchase of seedlings from the plantation and the loss of income during the time it takes for young palms to become productive, they point out that the oil palm company running the plantation is actually providing a barrier to the maintenance of palm oil productivity by smallholders. They argue that by formulating more innovative and sustainable smallholder policies, such as intercropping or other forms of diversification, the palm oil industry would be more likely to succeed in their aim to halt or reverse the decline of palm oil production on smallholder land. More advice to benefit smallholder palm oil production has been provided by the IIED to support the work of the Round Table on Sustainable Palm Oil [23]. These studies suggest that it should be possible for smallholders to experience positive social impacts from the expansion of the production of palm oil, but that examples where this happens are still rare and need to be delivered through strong and targeted policy

[2] See for example the following media reports (last accessed 27/10/2010: BBC 12 September 2010). Brazil rescues farm workers from slave-like conditions http://www.bbc.co.uk/news/world_latin-america-11274877 BBC 19 July 2004. Brazil 'slavery' damned by report http://news.bbc.co.uk/1/hi/world/americas/3908271.stm

interventions in favour of the smallholders. On a somewhat similar note, a good example of multifunctional use of Jatropha by rural communities has been reported in Africa [24]. However also here the positive social impacts appear to relate only to small-scale production by rural communities. This will come as no surprise if we take a historically informed view of the agricultural industry in the era of globalization [25].

5. Discussion

The above sections have shown that different social impacts can be anticipated dependent on the scale and location of production and consumption. The choice and consequences of these three aspects are discussed in turn.

5.1. *Motivation for production*

Since the earliest days of the car age when Alfred Diesel designed his engine to run on peanut oil, and Henry Ford designed his T-model to run on ethanol as well as petrol, there have been only a handful of localised situations where crude oil was not sufficiently cheap and plentiful and had to make way for 'alternative' fuels. The large scale production of synthetic oil from coal in Nazi Germany in the latter part of the Second World War and in Apartheid South Africa during the height of the economic boycott provides two notable examples of such unique situations. A third example is provided by Brazil's military leaders' decision, informed by the oil crisis of 1973, to switch to alcohol from sugarcane. Apart from the unsavoury human rights records of these anti-democratic regimes, these three events also share a common historical context of international conflict and strong, top-down government intervention to support the national production of alternative fuels. The current drive amongst a wide range of national governments towards transport biofuels is also led by strong top-down government interventions. It lies beyond the scope of this paper to discuss for different countries the potential implications for social policy that might arise from a reduced national dependency on the import of foreign oil and higher domestic fuel prices or subsidies for liquid biofuel production. However the key message emerging from the examples of historic and current national policies on alternative fuels is that the key driver is neither environmental nor social. Whatever the formal justification is for government policies to stimulate the production of liquid biofuels, the explicit support for large scale production suggests that energy security is a far more important motive. In the case of subsidised domestic production, the political 'tradition' of supporting the farming sector also plays a role, although it is ironic that that tradition was often justified on the basis of food security – the competitor of energy security in the case of first generation biofuels.

5.2. *Motivation for consumption*

The production and use of biofuels is never carbon neutral. At best it is less carbon intensive than the petroleum products it displaces. Cheaper fuel results in greater overall use of fuel, i.e. it results in a greater overall environmental impact. Hence the drive to make biofuels cheaper is at odds with the overall environmental aim of reducing greenhouse gas emissions. This environmental aim is better served by making fossil fuels more expensive, e.g. through a carbon tax. When the end-use is considered in a more holistic manner, questions can and should be asked about the social impacts of car use in general. The car economy has a wide range of negative impacts on public health (air pollution, noise pollution, accidents), the increased and wasteful usage of energy and space (tarmac, parking space, suburbia) and even social relations (congestion and commuting time, sole commuters, road rage, isolation and exclusion of those who don't have a car) [26].

5.3. *Choice of scale*

It can of course be argued that, at some scale at least, increased automobility will have positive social impacts. Ambulances save lives, and so do heavy goods vehicles and other mechanised forms of transport who evacuate people at risk of natural disasters or bring in supplies into areas devastated by floods, droughts etc. In short, cheap transport biofuels are mainly likely to yield social benefits in poor countries where private access to cars is very limited. Cheaper fuel would potentially help the government and NGOs to spend less of their budget on fuel and free up money to extend the social services they provide.

From a social impacts perspective, another benefit of small-scale production is that it links the producer with the consumer in repeated exchanges that include both financial transactions and social interactions. This social scale reduces the likelihood that negative social impacts will go unnoticed or unmitigated. Large scale and globalised production models are much more likely to result in negative social impacts, caused or exacerbated by the geographical, cultural and power divide between the governments and large companies who are driving this agenda forwards and the individuals and communities affected 'on the ground'. In theory sustainability certification systems can be implemented to create channels for accounting, reporting and mitigating these impacts. However for these systems to work, they not only have to be extensively (and thus expensively) designed and policed but they also have to be driven or endorsed by consumer demand. With so many environmental, conservation and development NGOs vocally opposed to the current and planned developments for biofuel production, it is difficult to see who could sway public opinion to buy into the belief that driving their cars on cane sugar ethanol or palm oil biodiesel amounts to an act of ethical consumerism.

In addition to questions about the desirability of the product, or the robustness of the certification process and its policing, important questions can also be asked about who manages the certification process and to what extent is its design and management (as it evolves over time) truly sympathetic to the needs of the poor (est). Experiences with certification to date suggests that the threshold for entering is too high for the poorest farmers, and this gap may further increase with the scaling up of production [27–29].

6. Conclusions

This paper provides a critical examination of some of the social impacts associated with the current 'fashion for transport

biofuels' [30]. We conclude that the production of transport biofuels could bring positive social impacts, but that these are very unlikely to emerge as automatic by-products of the large scale production of bioethanol or biodiesel. Just as with positive environmental impacts of transport biofuels, positive social impacts of transport biofuels cannot be expected to emerge without strict regulation of the entire supply chain. It is argued that the safeguarding of positive social impacts will often require a shift in thinking whereby the production of liquid biofuels is seen as a by-product or a co-product of a more multifunctional production process that has the delivery of local socio-economic or environmental benefits as its main objective. In (post) industrial societies, the social benefits of producing transport biofuels are highly debatable because it would do nothing to alleviate the social problems caused by excessive car usage, the farming community is small, not particularly poor and has been heavily subsidised for decades. Wealthy consumers in these societies can deliver social benefits in developing countries by paying a premium price for socially and environmentally certified biofuels imported from developing countries. This practice would be very similar to that of organic and fair trade certification, and would raise the same range of moral, technical and administrative issues. Such certified premium products would have to be specially marketed and branded and a widespread availability of non-certified (i.e. socially and environmentally controversial) liquid biofuels could cause significant damage to the brand and reduce demand. The way in which liquid biofuels are currently promoted in industrial countries is very much large scale, import-oriented and supply-led; an approach which is very likely to yield negative social and environmental impacts in producer countries. International certification for minimal social and environmental standards could make a difference in theory but in practice it would require a monitoring and policing effort that would be unparalleled in the history of international trade of commodities of such (relatively high) volume and (relatively low) value. The costs of monitoring and policing are much lower for localised small-scale systems, because there is less complexity, more individual contact and social contract between stakeholders and also smaller overall profits to tempt cheaters or test the strength of corporate ethics. Current models for the utilisation of waste cooking oil as a transport fuel provide an example that is often characterised by a strong and personal relationship between the consumer and small-scale producer and the waste oil providers. Such relationships between producer and consumer that are built on personal trust and shared values (e.g. regarding environmentalism or regarding a sense of self sufficiency and independence from distrusted, feared or disliked powers such as big oil multinationals or certain oil exporting countries), are much more likely to provide positive social impacts to both parties involved in this exchange. More empirical research (i.e. fieldwork) would be required to assess the social impacts of small-scale biofuel production systems in rural areas of developing countries, but it could be envisaged that such systems, when developed ('bottom-up') by the people involved, would share many of the characteristics of the small-scale and localised systems in developed countries and thus potentially yield similar positive social impacts. In fact these systems could potentially yield additional social benefits if they were to extend (new) energy services to rural people who had not enjoyed these before.

Going back to the generic claims of sustainability made by some academic supporters of biofuels, this paper has highlighted a fundamental contradiction. It is indeed true that plants can and do have multiple functions and uses, but these are mostly valued by those whose livelihoods (subsistence farmers) or life styles (e.g. the eco village movement, green consumers) are closely aligned to nature. Commercial production systems on the other hand, tend to maximise only the best-selling product or ecosystem service, at the expense of others. If biofuels are to be produced sustainably and in substantial quantities, there is a need for market interventions by the government. Governments may pays lip service to various social and environmental objectives but from the details in the design of support measures for biofuels, it is possible to observe that security of supply is a far greater priority. Clearly policies that are designed for the purpose of security of energy supply cannot realistically be expected to yield high social or environmental benefits, and certainly not abroad.

Acknowledgements

Dan would like to thank the UK Economics and Social Sciences Research Council (ESRC) for a fellowship grant on social enterprises and renewable energy (project reference RES-152-27-0004).

REFERENCES

[1] Van der Horst D. UK biomass energy since 1990; the mismatch between project types and policy objectives. Energy Policy 2005;33(5):705–16.
[2] Kandpal JB, Madan M. *Jatropha curcas*: a renewable source of energy for meeting future energy needs. Renew Energy 1995; 6(2):159–60.
[3] Sumathi S, Chai SP, Mohamed AR. Utilization of oil palm as a source of renewable energy in Malaysia. Renew Sustain Energy Rev 2008;12(9):2404–21.
[4] The ecologist. Special report on the biofuel revolution. London; March issue 2007.
[5] Seedling. Agrofuels special issue. Barcelona: Grain; July 2007.
[6] IOCPG. Principles and guidelines for social impact assessment in the USA. The Interorganisational committee on principles and guidelines for social impact assessment. Impact Assess Proj Apprais 2003;21(3):231–50.
[7] World Bank. Social analysis sourcebook: incorporating social dimensions into bank-supported projects. Washington DC: World Bank; 2002.
[8] World Bank. A user's guide to poverty and social impact analysis. Washington DC: World Bank; 2003.
[9] Vanclay F. Principles for social impact assessment: a critical comparison between the international and US documents. Environ Impact Assess Rev 2006;26:3–14.
[10] Cole MA. Air Pollution and 'Dirty' industries: how and why does the composition of manufacturing output change with economic development? Environ Resource Econ 2000;17(1): 109–23.

[11] Cole MA, Elliot RJ. FDI and the capital intensity of 'dirty sectors': a missing piece of the pollution haven puzzle. Rev Develop Econ 2005;9(4):530–48.
[12] Van der Horst D, Evans J. Carbon claims and energy landscapes: exploring the political ecology of biomass. Landsc Res 2010;35(2):173–93.
[13] Vermeylen S. Land rights and the legacy of colonialism. In: Weinberg R, Chennells R, editors. Community consent and benefit-sharing: learning lessons from the San Hoodia case. Berlin: Springer Verlag; 2008.
[14] Frayne B. Migration and urban survival strategies in Windhoek, Namibia. Geoforum 2004;35(4):489–505.
[15] Costanza R, d'Arge R, de Groot R, Farber S, Grasso M, Hannon B, et al. The value of the world's ecosystem services and natural capital. Ecol Econ 1998;25(1):3–15.
[16] Gren I, Groth K, Sylven M. Economic values of Danube floodplain. J Environ Manage 1995;45:333–45.
[17] Guo Z, Xiao X, Gan Y, Zheng Y. Ecosystem functions, services and their values – a case study in Xingshan County of China. Ecol Econ 2001;38:141–54.
[18] Browne AW, Harris PJC, Hofny-Collins AH, Pasiecznik N, Wallace RR. Organic production and ethical trade: definition, practice and links. Food Policy 2000;25:69–89.
[19] Chambers R, Conway G. Sustainable rural livelihoods: practical concepts for the 21st century, vol. 296. Brighton, UK: Institute of Development Studies; 1992. IDS discussion paper.
[20] Tauli-Corpuz V, Tamang P. Oil palm and other commercial tree plantations, monocropping: impacts on indigenous peoples' land tenure and resource management systems and livelihoods. New York: United Nations Permanent Forum on Indigenous Issues (UNPFII); 2007. E/C.19/2007/CRP.6.
[21] Openshaw K. A review of *Jatropha curcas*: an oil plant of unfulfilled promise. Biomass Bioenergy 2000;19:1–15.
[22] Koczberski G, Curry GN. Making a living: land pressures and changing livelihood strategies among oil palm settlers in Papua New Guinea. Agric Syst 2005;85:324–39.
[23] Vermeulen S, Goad N. Towards better practice in smallholder palm oil production. IIED; 2006.
[24] Henning RK. Using the indigenous knowledge of Jatropha. World Bank; August 2002. IKNotes 47.
[25] Friedmann H. From colonialism to green capitalism: social movements and emergence of food regimes. Res Rural Sociol Dev 2006;11:227–64.
[26] Freund P, Martin G. The ecology of the automobile. Montreal: Black Rose Books; 1993.
[27] Gomex Tovar L, Martin L, Gomez Cruz MA, Mutersbaugh T. Certified organic agriculture in Mexico: market connections and certification practices in large and small producers. J Rural Stud 2005;21:461–74.
[28] Goodman MK. Reading fair trade; political ecological imaginary and the moral economy of fair trade foods. Polit Geogr 2004;23:891–915.
[29] Renard MC. Quality certification, regulation and power in fair trade. J Rural Stud 2005;21:419–31.
[30] Clift R, Mulugetta Y. A plea for common sense (and biomass). Chem Eng (TCE) October 2007;(issue):24–6.